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Mark scheme

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Key to mark scheme abbreviations

M	Mark is for method
m	Mark is dependent on one or more M marks and is for method
A	Mark is dependent on M or m marks and is for accuracy
B	Mark is independent of M or m marks and is for method and accuracy
E	Mark is for explanation
√ or ft	Follow through from previous incorrect result
CAO	Correct answer only
CSO	Correct solution only
AWFW	Anything which falls within
AWRT	Anything which rounds to
ACF	Any correct form
AG	Answer given
SC	Special case
oe	Or equivalent
A2, 1	2 or 1 (or 0) accuracy marks
–*x* EE	Deduct *x* marks for each error
NMS	No method shown
PI	Possibly implied
SCA	Substantially correct approach
sf	Significant figure(s)
dp	Decimal place(s)

Q	Answer	Marks	Comments
1(a)	$z_{\text{crit}} = 2.326$	B1	Condone 2.3263
		1	

Q	Answer	Marks	Comments
1(b)	$z = \dfrac{\lvert 10.1 - 9.5 \rvert}{\sqrt{\dfrac{4.2}{100} + \dfrac{4.8}{120}}}$	M1 M1	**M1**: Correct numerator (positive or negative) **PI** by correct [±] z **M1**: Correct denominator **PI** by correct [±] z
	$= 2.095[29..]$	A1	Exact value is $\dfrac{6\sqrt{205}}{41}$ Allow negative value $p = 0.01807$ If **M0 M0** awarded, allow **SC1** for use of t-statistic with value [±] 2.083
	$2.095 < 2.326$ Therefore do not reject H_0	B1ft	**ft** their critical value and their test statistic with the corresponding correct conclusion
		4	

Q	Answer	Marks	Comments
1(c)	Sample is large enough to approximate with a normal distribution	B1	Any correct assumption, such as correct reference to central limit theorem
		1	

	Question 1 Total	**6**	

Q	Answer	Marks	Comments
2	H_0: There is no association between location of a supporter and their predicted winning team H_1: There is an association between location of a supporter and their predicted winning team	B1	Both H_0 and H_1 correct
		M1	At least three correct
		A1	All correct
	$\sum \frac{(O-E)^2}{E} = \frac{(140-"127.2")^2}{"127.2"} + \frac{(101-"89.4")^2}{"89.4"} + \frac{(59-"83.4")^2}{"83.4"} + \frac{(72-"84.8")^2}{"84.8"} + \frac{(48-"59.6")^2}{"59.6"} + \frac{(80-"55.6")^2}{"55.6"}$	m1	**PI**, six terms with at least three correct
	$= 24.8295121...$	A1	**AWRT** 24.8, $p-$value of 4.06×10^{-6}
	dof $= v = (3-1)(2-1) = 2$	B1	**PI**
	$\chi^2(0.99) = 9.210$	B1	Comparison may use p value, $0.01 > 4.06\times10^{-6}$
	$24.82\ldots > 9.210$, Therefore reject H_0		
	Sufficient evidence to suggest there is an association between the location of a supporter inside or outside the host country and their predicted winning team	A1ft	Final statement must be consistent with their test statistic
		8	

	Question 2 Total	**8**	

Q	Answer	Marks	Comments

Brazil	Germany	Other
Host country	**127.2**	**89.4**	**83.4**
Outside the host country	**84.8**	**59.6**	**55.6**

3(a)	$z = 1.96$ $1.96 \times \frac{\sqrt{6.25}}{\sqrt{n}} = \frac{1.4}{2}$ $n = 49$	**B1** **M1** **A1**	**PI** Set up correct equation with their z value
		3	

Q	Answer	Marks	Comments
3(b)(i)	$(5.6, 7.0)$	**B1**	Condone 7 instead of 7.0
		1	

Q	Answer	Marks	Comments
3(b)(ii)	5.4 is not in the confidence interval, therefore reject H_0	**B1ft**	**ft** with their interval
		1	

Q	Answer	Marks	Comments
3(c)	$P(4.7 < \overline{X} < 6.1 \mid \mu = 5.7)$ $\frac{\sqrt{6.25}}{\sqrt{49}}$ or $\frac{5}{14}$ or $0.357[14\ldots]$ $P(4.7 < \overline{X} < 6.1 \mid \mu = 5.7) = 0.866[08\ldots]$ $1 - 0.866\ldots$ $= 0.134$	**M1** **B1** **A1** **A1**	**oe** **PI** Use of standard error **PI** by use of standardisation of z value or $z_1 = -2.8$ or $z_2 = 1.12$ or correct answer **CAO**
		4	

	Question 3 Total	**9**	

Q	Answer	Marks	Comments
4(a)(i)	$\mathrm{E}(\bar{X})\left[=\frac{1}{n}n\mu\right]=\mu$ $\mathrm{E}(\bar{Y})\left[=\frac{1}{m}m\mu\right]=\mu$	M1	Both expectations equal to μ seen or used
	$\mathrm{E}(T)=\frac{n}{n+m}\times\mu+\frac{m}{n+m}\times\mu=\mu$	A1	
	T is an unbiased estimator of μ	E1	
		3	

Q	Answer	Marks	Comments
4(a)(ii)	$\mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}$ and $\mathrm{Var}(\bar{Y})=\frac{\sigma^2}{m}$	M1	Both variances seen or used
	$\mathrm{Var}(T)=\left(\frac{n}{n+m}\right)^2\times\frac{\sigma^2}{n}+\left(\frac{m}{n+m}\right)^2\times\frac{\sigma^2}{m}$		Must see this line of working
	$\mathrm{Var}(T)=\frac{\sigma^2}{n+m}$	A1	**AG**
		2	

Q	Answer	Marks	Comments
4(a)(iii)	$\mathrm{Var}(T)\to 0$ as $n\to\infty$ and $m\to\infty$	M1	**oe** $\mathrm{Var}(\bar{X})\to 0$ as $n\to\infty$ $\mathrm{Var}(\bar{Y})\to 0$ as $m\to\infty$
	Estimator is consistent for either sample	A1	**AG** Conclusion must be stated
		2	

Q	Answer	Marks	Comments
4(b)(i)	$\mathrm{E}\left(S_x^2\right)=\sigma^2, \quad \mathrm{E}\left(S_y^2\right)=\sigma^2$	B1	**B1** for either expectation seen or used
	$E(V)=\dfrac{n-1}{n+m-2}\sigma^2+k\sigma^2$		
	$\dfrac{n-1}{n+m-2}\sigma^2+k\sigma^2=\sigma^2$	M1	Must see intermediate working
	$k\sigma^2=\sigma^2-\dfrac{n-1}{n+m-2}\sigma^2$		
	$k=\dfrac{m-1}{n+m-2}$	A1	**AG** Be convinced
		3	

Q	Answer	Marks	Comments
4(b)(ii)	Pooled [estimate of population] variance	B1	**oe** combined variance, composite variance
		1	

Q	Answer	Marks	Comments
4(b)(iii)	Test for the difference of two means [when the variances are unknown and may be assumed equal]	B1	Allow t-test [with 2 samples]
		1	

	Question 4 Total	**12**	

Q	Answer	Marks	Comments
5	$H_0 : \mu_{2021} = \mu_{2020}$ $H_1 : \mu_{2021} > \mu_{2020}$	B1	Both hypotheses correct
		B1	Allow 1 mistake Allow negative of table values
	dof $= v = 7$	B1	PI
	$[\text{critical value, } t_7 =]$ 1.415	B1	
	$\bar{d} = 1.275$	B1	
	$s^2 = \frac{1}{8-1}\left(\sum d^2 - 8\bar{d}^2\right)$	M1	PI Attempt at variance formula; allow one slip. Implied by correct answer $\sum d^2 = 51.68$
	$= 5.525$	A1	Accept $s =$ **AWRT** 2.35
	$t = \frac{\bar{d}}{s/\sqrt{8}} = \frac{1.275}{\sqrt{\frac{5.525}{8}}}$	M1	Using their mean and variance
	$t = +1.534[22\ldots]$	A1	p-value 0.0844
	$1.534 > 1.415$, therefore reject H_0	A1ft	Comparison may use p value $0.10 > 0.0844$
	Sufficient evidence to suggest the percentage of trains that arrive at stations on time has increased	E1	Correct conclusion based on their comparison
		11	

	Question 5 Total	11	

Q	Answer	Marks	Comments
6	$H_0 : \sigma_A^2 = \sigma_B^2$ $H_1 : \sigma_A^2 < \sigma_B^2$	B1	Both hypotheses needed Correct direction of inequality needed for H_1 Allow hypotheses in terms of σ instead of σ^2
	dof $v_A = 11, v_B = 11$	B1	**PI**
	$F_{11,11}$ at 99% $= 4.462$	B1	
	$s_A^2 = \frac{1}{12-1}\left(19498 - \frac{482^2}{12}\right)$ $[= 12.51\ldots]$ $s_B^2 = \frac{1}{12-1}\left(19531 - \frac{477^2}{12}\right)$ $[= 51.84\ldots]$	M1	**oe** $\frac{413}{33}$ **oe** $\frac{2281}{44}$
	$\frac{s_B^2}{s_A^2} = \frac{51.84\ldots}{12.51\ldots}$	M1	**oe** $\frac{6843}{1652}$ condone $\frac{s_A^2}{s_B^2} = \frac{12.51\ldots}{51.84\ldots}$
	$= 4.142$	A1	Correct value $p-$value of 8.3×10^{-3}
	$4.462 > 4.142$ Do not reject H_0	B1ft	$8.3 \times 10^{-3} < 0.01$ Allow 'accept H_0'
	Insufficient evidence to suggest less variance in the number of faulty chips in each box for Company A	E1	**oe** allow 'manufacturing process is better' or support for the company's claim is not justified
		8	

	Question 6 Total	8	

Q	Answer	Marks	Comments
7(a)	$\mathrm{M}_X(t)=\mathrm{E}\left(\mathrm{e}^{tX}\right)=(1-p)\mathrm{e}^0+p\mathrm{e}^t$	M1	
	$=(1-p)+p\mathrm{e}^t$	A1	If **M0** awarded, **SC1** for $(1-p)+p\mathrm{e}^t$
		2	

Q	Answer	Marks	Comments
7(b)	$M_S(t)=\left((1-p)+p\mathrm{e}^t\right)^n$	B1	
		1	

Q	Answer	Marks	Comments
7(c)(i)	$M_S'(t)=np\mathrm{e}^t\left((1-p)+p\mathrm{e}^t\right)^{n-1}$	M1	Allow one slip for differentiating the correct answer from **part (b)** If **part (b)** scored 0, then **M1** for correct first derivative from their **part (b)**
	$M_S'(0)=np\mathrm{e}^0\left((1-p)+p\mathrm{e}^0\right)^{n-1}$	m1	Their first derivative evaluated at $t=0$
	$\mathrm{E}(S)=M_S'(0)=np$	A1	**CSO**
		3	

Q	Answer	Marks	Comments
7(c)(ii)	$M_S''(t)=n(n-1)p^2\mathrm{e}^{2t}\left((1-p)+p\mathrm{e}^t\right)^{n-2}$ $+np\mathrm{e}^t\left((1-p)+p\mathrm{e}^t\right)^{n-1}$	M1	**M1** Correct second derivative, allow one slip
	$M_S''(0)=np(1-p)+n^2p^2$	A1	Correct second derivative evaluated at $t=0$
	$\mathrm{Var}(S)=M_S''(0)-\left(M_S'(0)\right)^2$	M1	**M1** for use of variance formula
	$=np(1-p)$	A1	**ACF, CSO**
		4	

	Question 7 Total	**10**	

Q	Answer	Marks	Comments
8(a)(i)	$\overline{X} = 2.1$ $p = \frac{2.1}{6} = 0.35$	B1	AG
		1	

Q	Answer	Marks	Comments
8(a)(ii)	94.20 0.74	B1 B1	
		2	

Q	Answer	Marks	Comments
8(a)(iii)	H_0: Binomial model is a suitable distribution H_1: Binomial model is not a suitable distribution	B1	
	dof $= \nu = 4$	B1	PI
	$\chi^2(0.975) = 11.143$	B1	
	$\sum \frac{(O-E)^2}{E} = \frac{(40-30.17)^2}{30.17} + \frac{(103-97.46)^2}{97.46} + \frac{(113-131.20)^2}{131.20} + \frac{(83-\text{“}94.20\text{”})^2}{\text{“}94.20\text{”}} + \frac{(45-38.04)^2}{38.04}$	M1	Attempt to calculate statistic
	$+ \frac{(16-\text{"}8.93\text{"})^2}{\text{"}8.93\text{"}}$	m1	Combines last two categories
	$= 14.25$	A1	**AWFW** [14.2, 14.3] p-value is 0.0141
	$14.25 > 11.143$, therefore reject H_0		Comparison may use p value, $0.025 > 0.0141$
	Insufficient evidence to suggest binomial model is a suitable distribution	E1	Both conclusion and comparison required
		7	

Q	Answer	Marks	Comments
8(b)(i)	$\left[400\times e^{-2.1}\times\frac{2.1^0}{0!}=\right]$ 48.98 $\left[400\times e^{-2.1}\times\frac{2.1^5}{5!}=\right]$ 16.67	B1 B1	
		2	

Q	Answer	Marks	Comments
8(b)(ii)	dof $=\nu=7-1-1=5$ $\chi^2(0.975)=12.833$ $8.41<12.833$, do not reject H_0 or sufficient evidence to suggest Poisson model is a suitable distribution	B1 B1 E1	**PI** Final 2 categories are not combined Allow 'accept H_0' Conclusion must be correct
		3	

Q	Answer	Marks	Comments
8(c)	The Poisson model is better and The Poisson model was not rejected [whereas the binomial model was rejected]	B1ft	Valid reason must be given. Allow explanation of lower χ^2 test statistic value for their choice of model
		1	

	Question 8 Total	**16**	